STAT 1110 – Summary Tables: Two Variables (Part I)*

Guiding Question

Is the question dealing with a difference in means (quantitative response variable) or a difference in proportions (categorical response variable)?

	Categorical Response Variable (Two Proportions: Chapter 5)	**Quantitative Response Variable** (Two Means: Chapter 6)
Graphical Summary	Segmented Bargraph, Mosaic Plot	Parallel Boxplots
Summary Measures	$\widehat{p}_1, \widehat{p}_2, n_1, n_2$	$\bar{x}_1, \bar{x}_2, n_1, n_2$
Summary Statistic	$\widehat{p}_1 - \widehat{p}_2$	$\bar{x}_1 - \bar{x}_2$
Parameter	$\pi_1 - \pi_2$	$\mu_1 - \mu_2$
Null Hypothesis	$H_0 : \pi_1 - \pi_2 = 0$ (i.e. $H_0 : \pi_1 = \pi_2$)	$H_0 : \mu_1 - \mu_2 = 0$ (i.e. $H_0 : \mu_1 = \mu_2$)
Alternative Hypothesis (Pick One)	H_A: $\pi_1 - \pi_2 > 0$ (i.e. H_A: $\pi_1 > \pi_2$) H_A: $\pi_1 - \pi_2 < 0$ (i.e. H_A: $\pi_1 < \pi_2$) H_A: $\pi_1 - \pi_2 \neq 0$ (i.e. H_A: $\pi_1 \neq \pi_2$)	H_A: $\mu_1 - \mu_2 > 0$ (i.e. H_A: $\mu_1 > \mu_2$) H_A: $\mu_1 - \mu_2 < 0$ (i.e. H_A: $\mu_1 < \mu_2$) H_A: $\mu_1 - \mu_2 \neq 0$ (i.e. H_A: $\mu_1 \neq \mu_2$)
Null Distribution	Distribution of difference in sample proportions, assuming the null hypothesis is true	Distribution of difference in sample means, assuming the null hypothesis is true
Validity Conditions for a Theory-Based Approach	Each value in the four interior cells of the two-way table is at least 10	Response variable has a symmetric distribution **–OR–** there are at least 20 observations in each group **and** the sample distributions are not heavily skewed
Standardized Statistic (Simulation-Based)	$z = \dfrac{(\widehat{p}_1 - \widehat{p}_2) - (\text{hypothesized mean of null distribution})}{SD_{NULL}}$	$t = \dfrac{(\bar{x}_1 - \bar{x}_2) - (\text{hypothesized mean of null distribution})}{SD_{NULL}}$
95% Confidence Interval (Using the 2SD Method)	$(\widehat{p}_1 - \widehat{p}_2) \pm 2 \cdot SD_{NULL}$ **–OR–** $(\widehat{p}_1 - \widehat{p}_2) \pm 2 \cdot \sqrt{\dfrac{\widehat{p}_1(1-\widehat{p}_1)}{n_1} + \dfrac{\widehat{p}_2(1-\widehat{p}_2)}{n_2}}$	$(\bar{x}_1 - \bar{x}_2) \pm 2 \cdot SD_{NULL}$ **–OR–** $(\bar{x}_1 - \bar{x}_2) \pm 2 \cdot \sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}$
SD_{NULL}	Obtained from the Multiple Proportions applet	Obtained from the Multiple Means applet
SE (Standard Error)	$SE(\widehat{p}_1 - \widehat{p}_2) = \sqrt{\dfrac{\widehat{p}_1(1-\widehat{p}_1)}{n_1} + \dfrac{\widehat{p}_2(1-\widehat{p}_2)}{n_2}}$	$SE(\bar{x}_1 - \bar{x}_2) = \sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}$

* ***Note:*** Paired data has been moved to Midterm 3.